Supplementary Information for: How do extreme weather events affect fishing operations? Knowledge engineering a risk analysis with Swedish Baltic Sea fishers

Supp. Inf. B: treatment of fisheries statistics available in European and Swedish respositories.

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Data sources by fishing style

Commercial fishers

Data for fishery dependent information for professional commercial fishers are available through the European Data Collection Franmework, which most recent update at the time of writing includes data until 2024. The dataset relevant for this study includes:

- *Fishery dependent information (FDI) for EU Fleets* (JRC, n.d.): provides fishing effort (dataset D1; fishing days) and catch (dataset D2; weight and value) data by country, area, fleet segment and target species
- *Annual Economic report on EU fishing fleet* (JRC 2025): provides economic overview (dataset D3; number of vessels, income, expenses) of european fishing fleets by country and fleet segment

The FDI data resolution allow to obtain information for the study area (Bothnian Sea - ICES area 27.3.D.30), while the AER is only available at country (Sweden) level. Different aggregation of the data requires to implement some simplifications or assumptions, hereby explained. Terminology adopted is consistent with definitions in use in the Data collection framework of the European Union ("Gear" 2026).

Small scale fishers

Small scale fishers catch and effort data correspond to vessels of size $<$ 12m, deploying passive gears and operating in the sub-region 27.3.D.30. This region includes the majority of small scale coastal fishers deploying gillnets in Sweden, especially for those targeting Herring (more than 50% of total fishers with GNS-SPF metier are registered in the study area). Economic data for the fleet segments, although refers to values at country level, are considered to be representative of the study area.

Pelagic trawlers

Pelagic correspond to vessels of size $>$ 24m operating pelagic nets (gear code: Trawl Midwater - TM) in the sub-region 27.3.D.30. Pelagic trawlers move between Swedish areas, and the same fleet can report catches in several divisions. Economic data at country level are therefore an appropriate aggregation. However, given the very small number of active vessels, Sweden does not provide economic data at fleet segment level for confidentiality reasons (JRC 2025). Economic data are provided for the aggregated Large Scale fleet, which includes vessels $>$ 24m operating demersal tralwers, pelagic trawlers and purse seiners. On average, over the period 2020-2025 40 % of landed value.

Recreational fishers

Dataset for recreational fishers is available through Sweden statistic reports (SWAM 2024), which most recent update at the time of writing includes data until 2024. The dataset relevant for this study are the following tables (table number ios reported as it appears in Swam):

- Table 5: target species in the sea in 2024
- Table 6: primary and secondary gear used in recreational fishing in 2024
- Table 8: Catches in recreational fishing in the sea in 2024

- Table 9: Expenditures in recreational fishing in 2024, million SEK

Dataset ID	Dataset_name	Description
D1	fdi_effort_SWE.csv	Fishing effort for commercial fisheries from FDI data call (JRC, n.d.)
D2	fdi_catch_SWE.csv	Catches for commercial fisheries from FDI data call (JRC, n.d.)
D3	fdi_SWE.csv	Economic overview (number of vessels, income, expenses) of european fishing fleets (JRC 2025)
D4	rec_gears_SWE.csv	Thousand of practitioners reported for recreational fishing technique in Sweden
D5	rec_catch_SWE.csv	Thousand of practitioners reported for target species in Sweden
D6	rec_economics_SWE.csv	Economic and fishing effort data

Fishing gears

Small-scale coastal fishing

We analysed dataset D1, subset to ICES area 27.3.D.30, Swedish country code, vessels categories VL0812 and VL0008 to identify the mostly used métiers by the Small-scale coastal fishing (ssf). A **métier** is defined as group of fishing activities targeting a similar species or assemblage of species, using similar gear (DCF 2021). Figure B.1 shows average total fishing days by métier over the available data (years 2008-2024). We select for consideration in our study the métiers (1) gillnets-freshwater species (GNS-FWS), (2) gillnets-small pelagic species (GNS-SPF), (3) pound nets- anadromous species (FPN-ANA). The sum of average fishing days for the three selected métiers accounts for 85% of total fishing days for ssf in our study area.

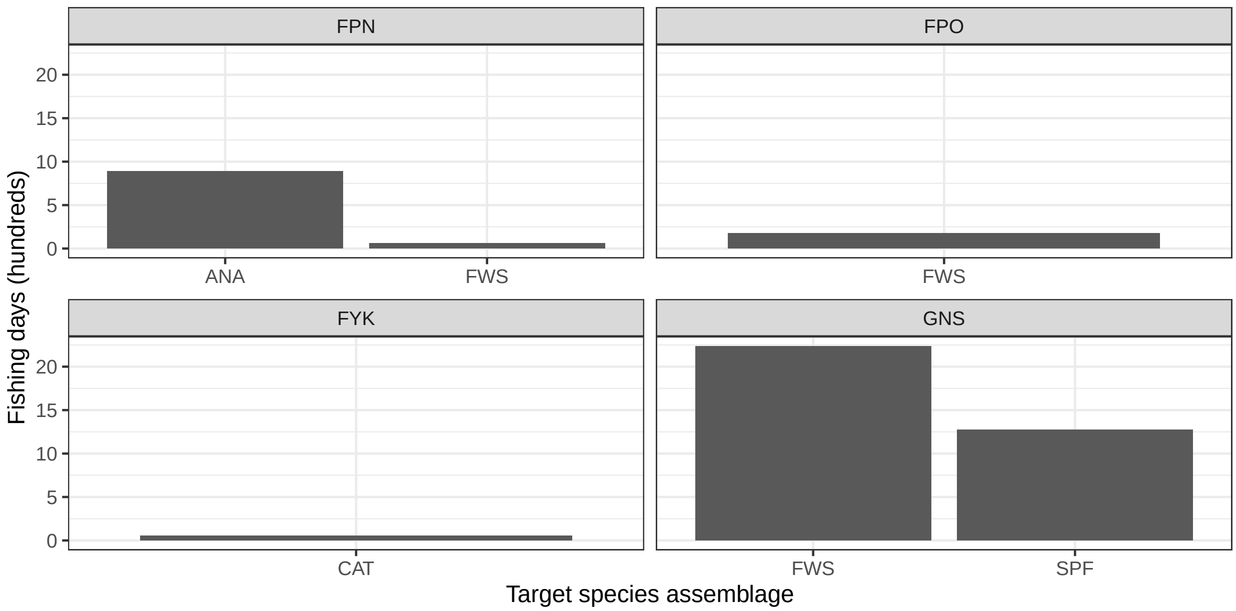

Figure B.1: Fishing days by metier, swedish small scale coastal fleet in Bothnian Sea. Panels represents gear type.

Pelagic trawling and large scale fleet

We analysed dataset D1, filtered to Swedish fleet, vessels categories VL2440 and VL40XX, to assess the fishing effort distribution by fishing technique in the Swedish large scale fleet. This information supports the interpretation of economic data, for which data are aggregated across fishing techniques. Figure B.2 shows that the fishing technique *Demersal Trawl Shelf* (DTS) and *Trawl Midwater* (TM) includes similar amount of fishing days, and togheter represents the vast majority of Swedish fishing effort. In the DTS category, pelagic trawling gears (OTM, OTT, PTM) dominates the fishing activity.

Recreational fishing

Dataset D4 shows the number of practitioners by recreational fishing method. Fishing methods based on hook and lines are dominating the distribution reported in Fig. B.3. Cages and nets are the two main techniques that do not base on hook and line. Cages are predominantly used in crustaceans fishery on the west coast of Sweden. The use of nets in popular in the Baltic Sea. For the purpose of this study we consider

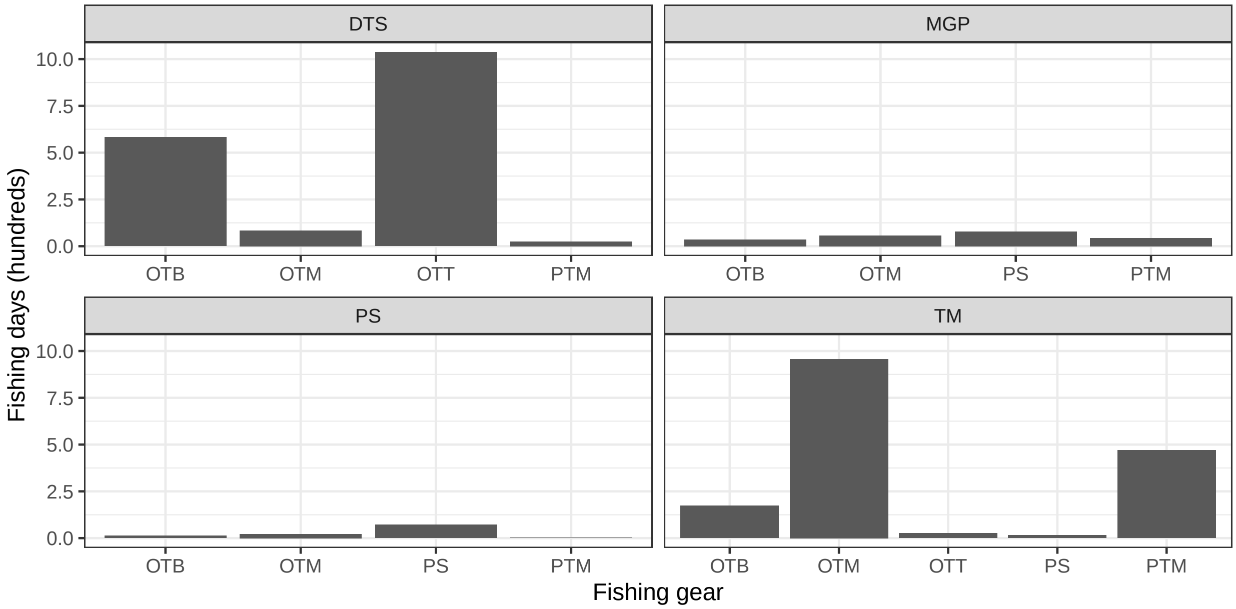

Figure B.2: Fishing days by gear and fishing technique, swedish large scale fleet. Panels represents fishing technique.

that only hook and line methods apply for the recreational fishing style, with no distinction among methods. Nets applies to the household fishing style.

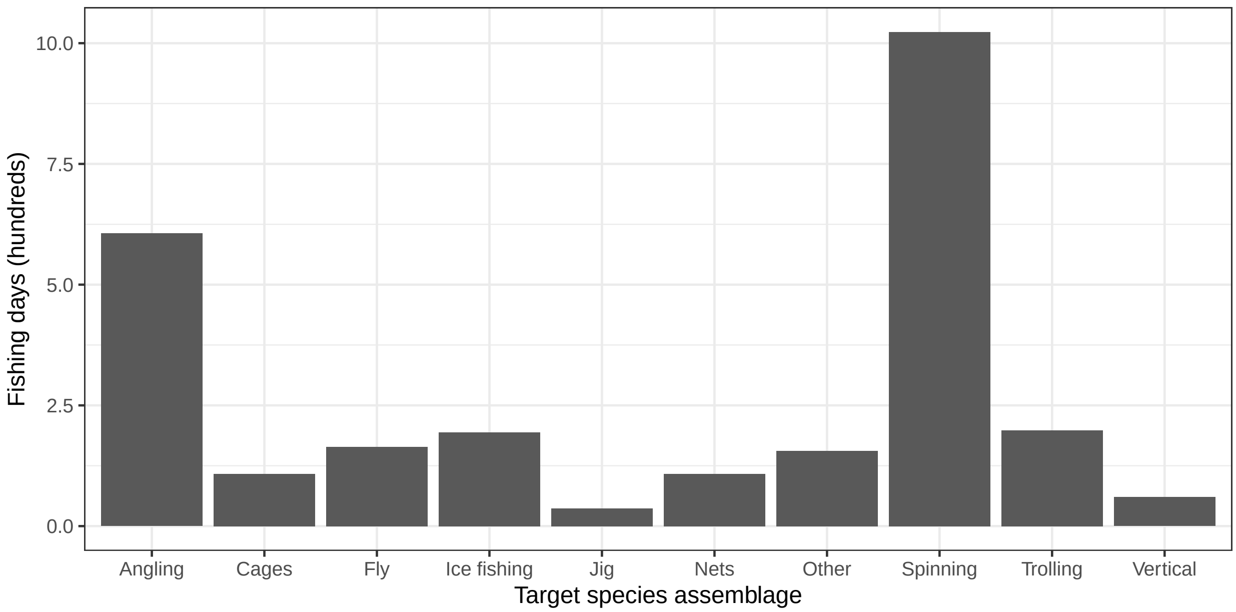

Figure B.3: Fishing days by fishing technique applied by swedish recreational fishers

Fishing target

Small-scale coastal fishing

We analysed dataset D2 to identify the catch composition associated to ssf métiers selected in the previous section. The dataset, filtered to Swedish fleet in ICES area 27.3.D.30, vessels categories VL0812 and VL0008, was subset to match the ssf métiers (1) gillnets-freshwater species (GNS-FWS), (2) gillnets-small pelagic species (GNS-SPF), (3) pound nets- anadromous species (FPN-ANA). Species scientific name were assigned according to ASFIS List of Species for Fishery Statistics Purposes (FAO 2026).

Catch data were aggregated by metier and averaged over years. We calculated the cumulative contribution of each species to a metier and we excluded species falling beyond the 95th percentile. Figure C.3 shows average tons by species and metier over the available data (years 2008-2024). The species mostly represented are Atlantic herring (target of gillnets for small pelagic fish), Atlantic salmon (main target of pound nets for anadromous species), European perch (main target of gillnets for freshwater species). Whitefish is also considered a relevant species as it significantly contribute to catches for FPN-ANA and GNS-FWS.

Pelagic trawling and large scale fleet

We analysed dataset D2 to evaluate the catch value associated with fishing gears deployed by Swedish large scale fleet, and then the catch composition of pelagic trawlers. The economic value of catches, filtered to Swedish fleet in ICES area 27.3.D.30, and vessel categories 'VL2440', 'VL40XX' was aggregated by metier and averaged over years. Figure B.5 shows that pelagic fleet (TM) dominates the distribution (58% of total), and the fishing gears Otter Trawl Midwater (OTM) contributes nearly to 37% of the entire landed value.

To explore fishing target of the pelagic trawl fleet in the study area, dataset D2 was subset to match the gear type "OTM", Swedish fleet in ICES area 27.3.D.30, and vessel categories 'VL2440', 'VL40XX'. Species scientific name were assigned according to ASFIS List of Species for Fishery Statistics Purposes (FAO 2026). We calculated the cumulative contribution of each species to a metier and we excluded species falling beyond the 99th percentile. Figure B.6 shows that Atlantic herring contributes for more than 97% of average catches. For pelagic trawlers we therefore consider Herring as the only relevant species.

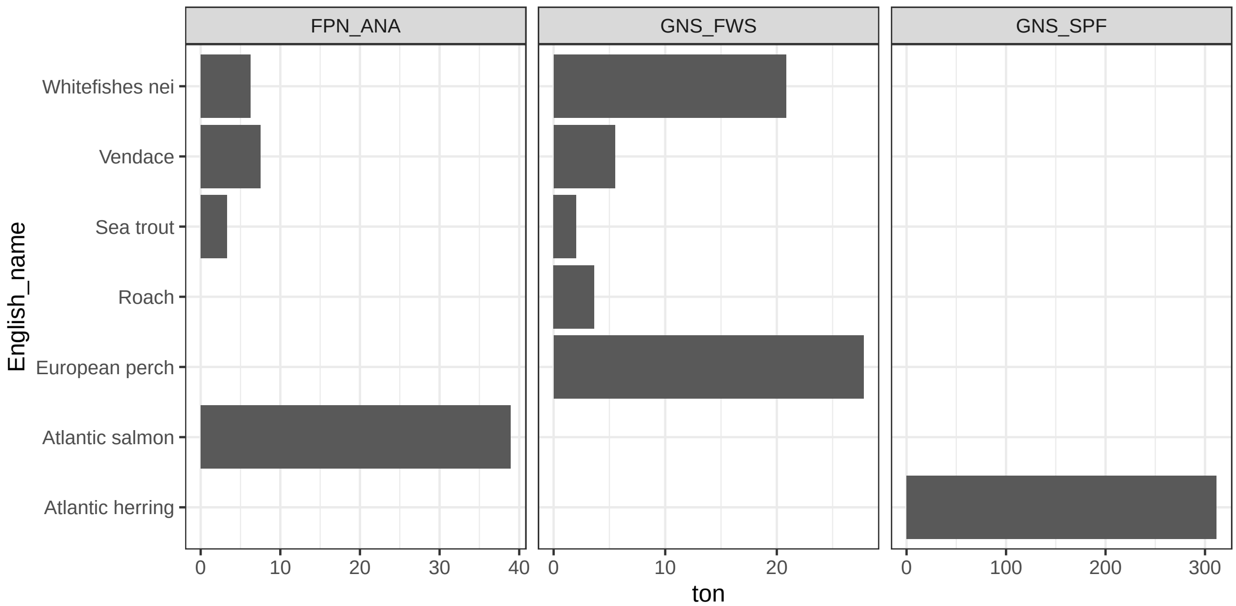

Figure B.4: Catches by metier, swedish small scale coastal fleet in Bothnian Sea (selection of metiers)

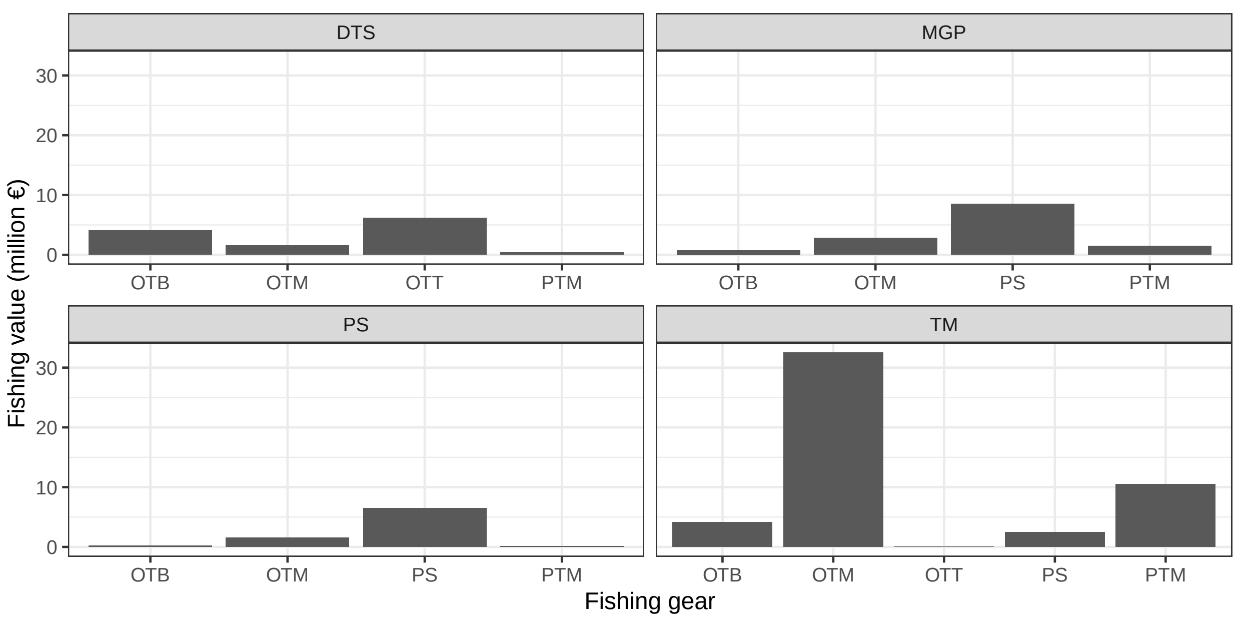

Figure B.5: Catches by gear and fishing method, swedish large scale fleet

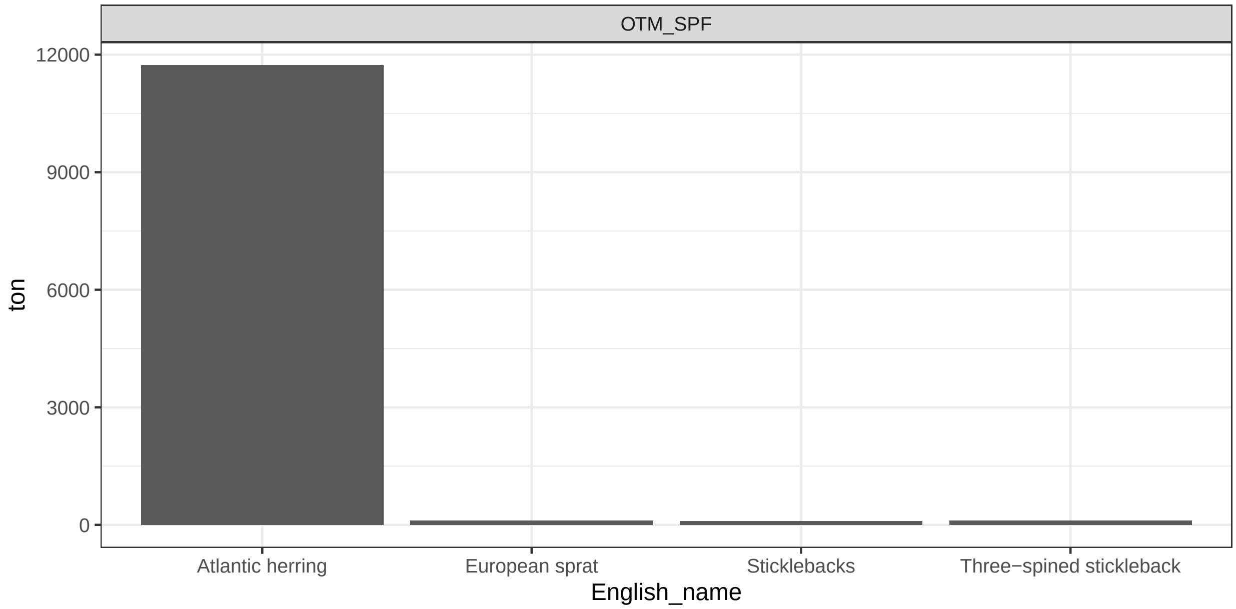

Figure B.6: Catches for pelagic trawlers, swedish fleet in Bothnian Sea

Recreational fishing

For recreational fishers we are interested in the distribution of number of practitioners by target species, in the Sea. Dataset D5 was subset to exclude species that are not available in the Baltic Sea. Fig. B.7 shows that Perch and pike dominates the distribution. We excluded some relevant species (Cod, Zander) for simplifyng the model. Whitefish was included, although not contributing substantially to recreational fishing target, because it is a shared resource with small scale fishers.

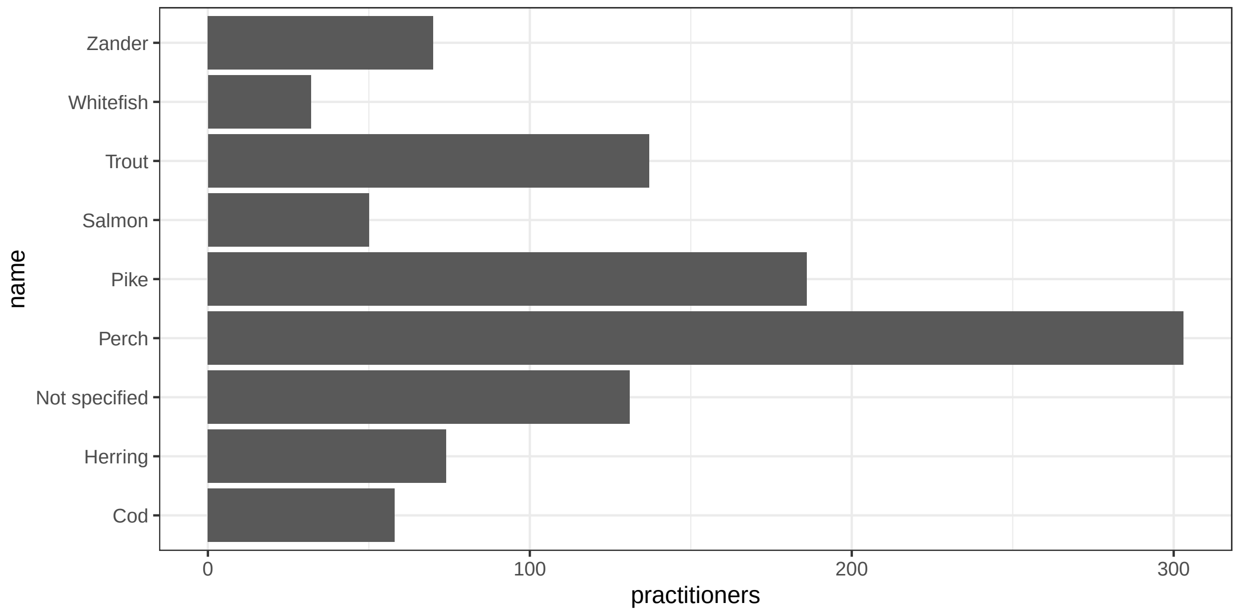

Figure B.7: Catches for recreational fishers

Economic data

Methods

The cost node reflects the economic impact of one single extreme weather event compared to annual average expenses. Cost categories to be used in the conditional probability table are: *negligible* (cost $< 0.05\%$); *low* ($0.05\% <$ cost $< 2.5\%$); *medium* ($2.5\% <$ cost $< 5\%$); *high* ($> 5\%$).

The conditional probability tables are estimated through the function `cost.cpt`, here below reported. The input needed are:

- expenses (mu and sd): annual total expenses, including maintenance, fuel, personnel. This value is anchored in literature, when available;
- fuel.cost (mu and sd): average cost for single fishing trip. This value is anchored in literature, when available;
- damage.small (lo and hi): self declared cost for repairing a minor damage;
- damage.large (lo and hi): self declared cost for repairing a large damage;
- damage.unc: self declared uncertainty on the cost for repairing damages;
- fuel.avg (lo and hi): self declared fuel consumption for single average fishing trip;
- fuel.large (lo and hi): self declared fuel consumption for single longer-than-average fishing trip;
- miss.catch (mu and sd): average gross economic value of catches from one fishing day
- economic.unc: self declared uncertainty associated with answer regarding expenses;
- fishing style: label describing the fishing style.

```r
cost.cpt=function(expenses.mu, expenses.sd,
                  fuel.cost.mu,fuel.cost.sd,
         damage.small.lo, damage.small.hi,damage.large.lo, damage.large.hi,
         damage.unc,
         fuel.avg.lo, fuel.avg.hi,
         fuel.large.lo, fuel.large.hi,
         miss.catch.mu,miss.catch.sd,
         economic.unc,
         f.style){

  # creating distributions
  ## distribution for annual expenses
  cost.annual=rtruncnorm(a=0, b=Inf,
                     mean=expenses.mu,
                     sd=expenses.sd, n=1000)

  ## distribution for minor damage
  cost.lo=rtruncnorm(a=damage.small.lo, b=damage.small.hi,
                     mean=(damage.small.hi-damage.small.lo)/2,
                     sd=damage.small.hi*damage.unc, n=1000)

  ## distribution for major damage
  cost.hi=rtruncnorm(a=damage.large.lo, b=damage.large.hi,
                     mean=(damage.large.hi-damage.large.lo)/2,
                     sd=damage.large.hi*damage.unc, n=1000)

  ## Distribution for avg travel Unit does not matter as long as it is the same of distribution for larg
  avg.fuel=(fuel.avg.hi+fuel.avg.lo)/2
  fuel.lo=rtruncnorm(a=fuel.avg.lo, b=fuel.avg.hi,
                     mean=avg.fuel,
                     sd=avg.fuel*economic.unc, n=1000)

  ## Distribution for large travel. Unit does not matter as long as it is the same of distribution for a
  lrg.fuel=(fuel.large.hi+fuel.large.lo)/2
  fuel.hi=rtruncnorm(a=fuel.large.lo, b=fuel.large.hi,
                     mean=lrg.fuel,
                     sd=lrg.fuel*economic.unc, n=1000)

  # Cost for fuel in extra travel. Multiplies proportion of large to avg travel by monetary cost.
  extra.fuel.cost=(fuel.hi/fuel.lo)*rtruncnorm(a=0, b=Inf,
                     mean=fuel.cost.mu,
                     sd=fuel.cost.sd, n=1000)

  # missing catches
  miss.catch=rtruncnorm(a=0, b=Inf, mean=miss.catch.mu, sd=miss.catch.sd, n=1000)

  # create data frame
  cost.df=data.frame(minor.no=cost.lo, major.no=cost.hi, no.travel=extra.fuel.cost)
  cost.df$minor.travel=cost.df$minor.no+cost.df$no.travel
  cost.df$major.travel=cost.df$major.no+cost.df$no.travel
  cost.df.2=cost.df+miss.catch
  names(cost.df.2)=paste(names(cost.df.2), 'miss', sep='.')
  cost.df.2$no.no.miss=miss.catch
```

```
  cost.df=cbind(cost.df, cost.df.2)
  cost.df[,]=apply(cost.df[,],2,function(x) x/cost.annual)
  cost.df.disc=cost.df
  cost.df.disc[,]=apply(cost.df.disc[,],2,
                      function(x) cut(x, c(-Inf, 0.005,0.025,0.05,Inf),
                                      labels=c('negligible','low','medium','high')))

  cost.df.disc$fishing_style=f.style
  cost.cpt=cost.df.disc%>%
  pivot_longer(-fishing_style, names_to = 'parent', values_to = 'state')%>%
  dplyr::group_by(fishing_style,parent, state)%>%
  tally()%>%
  dplyr::mutate(prob=n/sum(n))

  return(cost.cpt)
}

error.propagation=function(x.mu, x.er, y.mu, y.er){
  z.mu=x.mu/y.mu
  z.er=sqrt(((x.er/x.mu)^2)+((y.er/y.mu)^2))*z.mu
  return(data.frame(mu=z.mu, sd=z.er))
}
```

Small-scale coastal fishing

Economic information for small scale coastal fishers, available in dataset D3 (Table C.1), filtered for vessel length categories VL0010 and VL0812, fishing technologies FPO and DFN and Swedish country code. Cost of fuel per trip is calculated by dividing total fuel expenses by number of fishing trips. Annual spending is estimated by dividing total annual expenses by number of fishing vessels. Gross value of catches in a single fishing trip is obtained by dividing total annual value of catches by number of fishing trips. Other values are provided in the interviews.

Table B.2: Variables from interviews

id_I	short_description	range.lwr	range.upr	unit_range
2	minor_damage_cost	10000	50000	sek_unit
3	minor_damage_cost	1	50000	sek_year
7	minor_damage_cost	1000	50000	sek_unit
2	major_damage_cost	100000	500000	sek_unit
3	major_damage_cost	50000	200000	sek
7	major_damage_cost	50000	100000	sek_unit
3	fuel_trip	1500	2500	sek_day
3	extra_fuel	3000	5000	sek_day
7	extra_fuel	50	100	B liter_trip

Table B.3: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	1702	1567	thousand SEK	STECF_AER
Fuel cost	6670	8668	SEK	STECF_AER

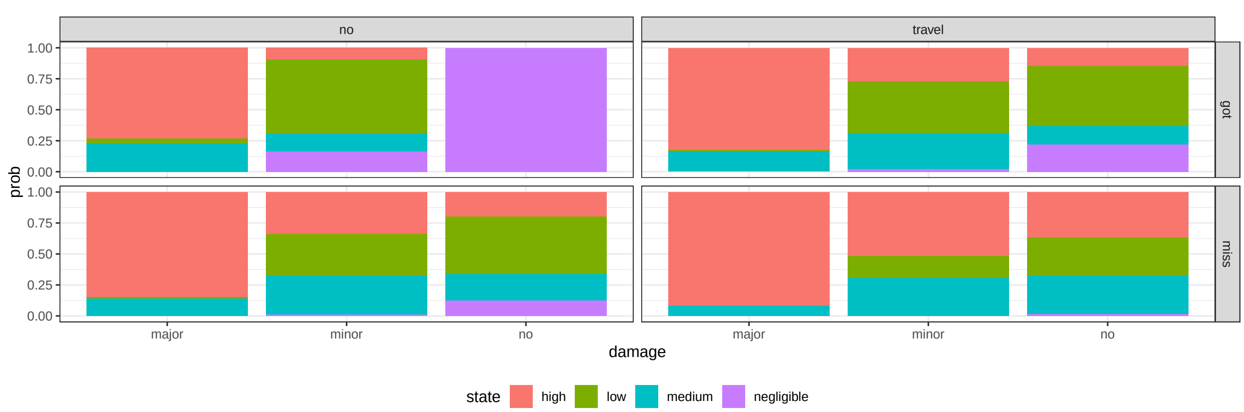

Figure B.8: CPT estimated for coastal fisherman

Pelagic trawling

Pelagic economic information is derived from dataset D3, filtered for vessel length categories VL2440 and VL40XX, and Sweden country code. Cost of fuel per trip is calculated by dividing total fuel expenses by number of fishing trips. Annual spending is estimated by dividing total annual expenses by number of fishing vessels. The cost of longer travel was not available, and it is based on the assumption that an average travel costs between 80% and 120% of the average fuel cost, while a longer travel is between 100% and 300% of average fuel cost. Other values are provided in the interviews.

[1] 4981303 5279201 5837720 5749069 5529277 6694086

Table B.4: Variables from interviews

id_I	short_description	range.lwr	range.upr	unit_range
6	minor_damage_cost	10000	50000	sek_unit
6	major_damage_cost	100000	500000	sek_unit
6	fuel_trip	600	650	liters_hour
6	extra_fuel	NA	NA	not_reported

Table B.5: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	50404	5643	thousand SEK	STECF_AER
Fuel cost	259142	117662	SEK	STECF_AER

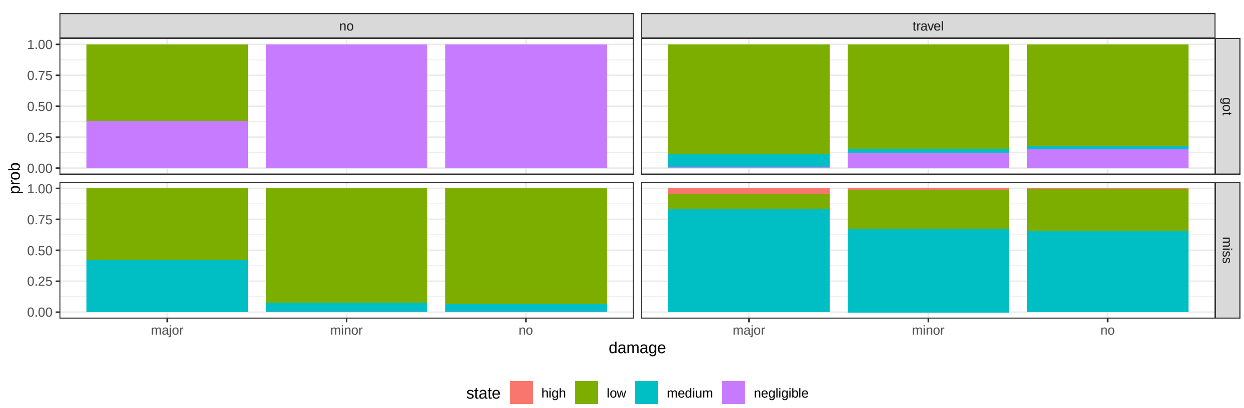

Figure B.9: CPT estimated for pelagic trawlers

Recreational fishing

Pelagic economic information is derived from dataset D6, representing the totality of recreational fisherman in Sweden, with no distinction between region or inland vs coastal fishing. Cost of fuel per trip is separately calculated for travel and for boat fuel. These two costs are then averaged. All the uncertainties (on costs and on number of practitioners) are accounted for by propagating errors. Annual spending is estimated by dividing total annual expenses by number of practitioners. Other values are provided in the interviews.

id_I	short_description	range.lwr	range.upr	unit_range
5	minor_damage_cost	10000	50000	sek_unit
5	major_damage_cost	50000	1000000	sek_unit
5	fuel_trip	30	50	B liter_trip
5	extra_fuel	20	50	B liter_trip

Table B.7: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	9213	3943	SEK	SWAM
Fuel cost	101	31	SEK	SWAM

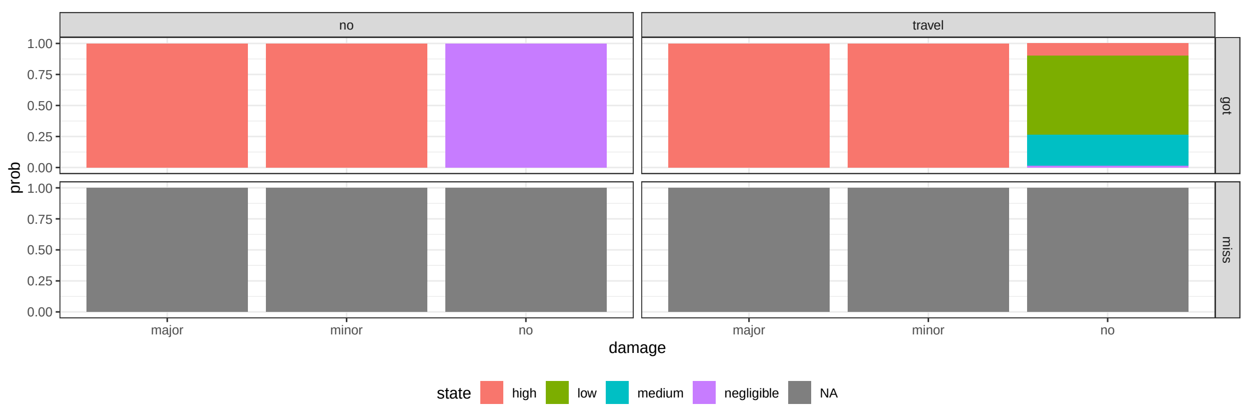

Figure B.10: CPT estimated for recreational fisherman

Fishing guides

No updated public information is available for fishing guide economic aspects in Sweden. All estimations base on interview results and our assumptions. Seasonal expenses are estimated by summing the declared maintenance costs and an estimation of fuel costs.

id_I	short_description	range.lwr	range.upr	unit_range
1	season_maint	50000	10000	sek_season
1	minor_damage_cost	10000	20000	sek_unit
1	major_damage_cost	20000	6000000	sek_unit
1	fuel_trip	1000	1000	sek_day
1	extra_fuel	500	500	sek_day

Table B.9: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	174898	14354	SEK	Our estimation based on interviews

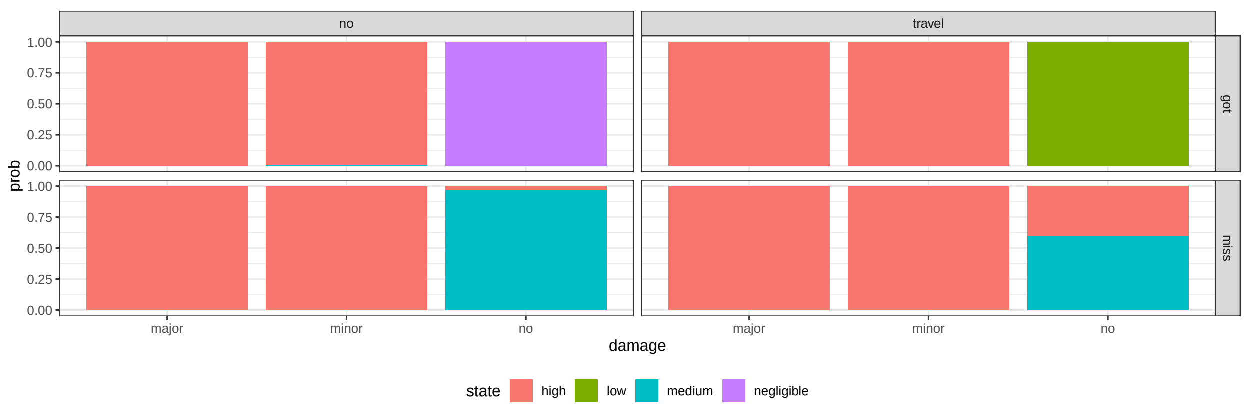

Figure B.11: CPT estimated for fishing guide

Household fishing

No updated public information is available for archipelago fishers economic aspects in Sweden. All estimations base on interview results and our assumptions. Seasonal expenses are estimated by summing the declared maintenance costs and an estimation of fuel costs.

id_I	**short_description**	**range.lwr**	**range.upr**	**unit_range**
4	season_maint	NA	5000	sek_season
4	minor_damage_cost	200	2500	sek_unit
4	major_damage_cost	2500	5000	sek_unit
4	fuel_trip	1000	6000	sek_season
4	extra_fuel	NA	NA	not_reported

Table B.11: Variables from literature

variable	**value_mean**	**value_sd**	**unit**	**source**
Annual expenses	8951	1700	SEK	Our estimation based on interviews

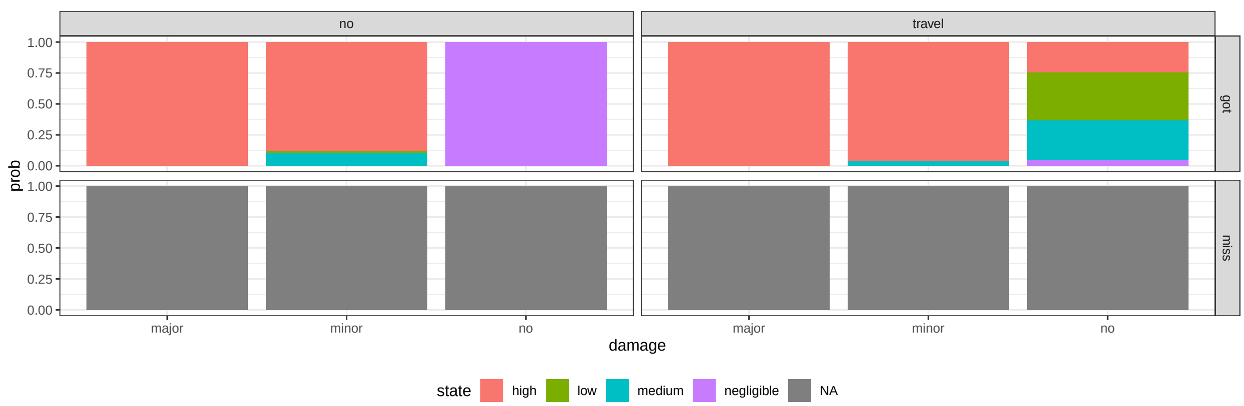

Figure B.12: CPT estimated for archipelago fisher

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